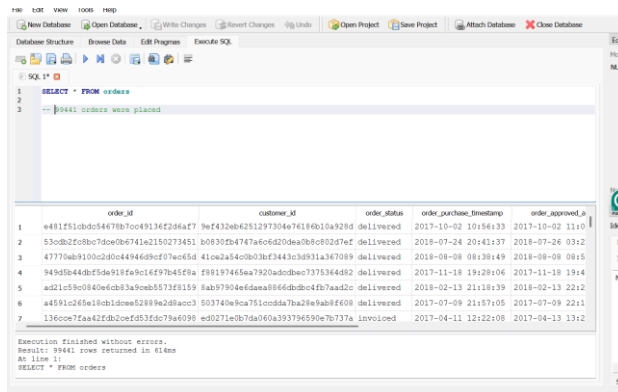


Question 1: How many orders were placed in total?

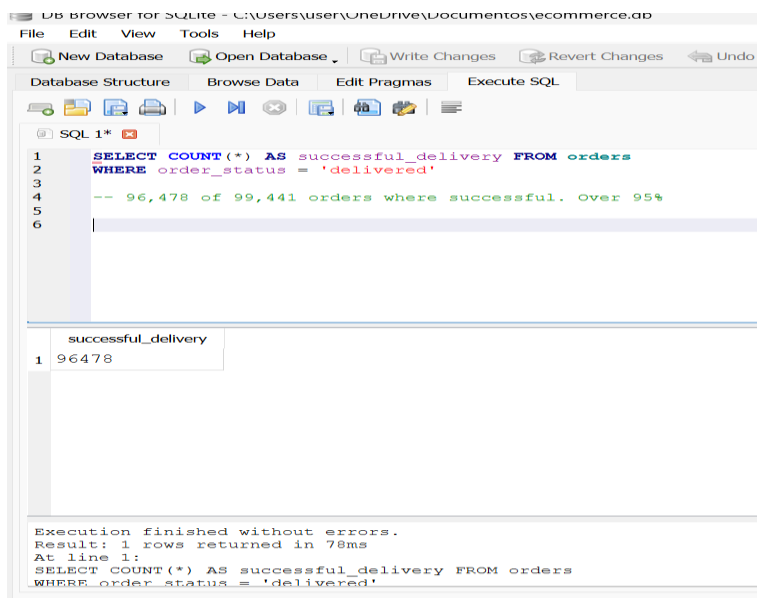


The screenshot shows a database management tool with a SQL editor and a results pane. The SQL editor contains the query: `SELECT * FROM orders`. The results pane displays a table with 7 columns: `order_id`, `customer_id`, `order_status`, `order_purchase_timestamp`, and `order_approved_at`. The table contains 7 rows of data, representing a subset of the 99,441 orders.

order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at
e481f51dbdc54678b70c49136f26aef7	9ef432ab6251297304e76194b15a928d	delivered	2017-10-02 10:16:13	2017-10-02 11:10
530db2f08bc7d0e0b6741e2150273451	b0930f54747a6c6d0dea08e092d7ef	delivered	2018-07-24 20:41:37	2018-07-26 03:12
47770ab9100c2d0c44946d8cf07ac65d	41ca2a54c0b03bf3443c3d931a367089	delivered	2018-08-08 08:38:49	2018-08-08 08:5
949db44df5d918f9c16f97b45f8a	f88197465ea7920ad0bec7375364482	delivered	2017-11-18 19:28:06	2017-11-18 19:4
ad21c59c0840efcb83a90ab5573f8159	8ab97904e6daa866dbcb4fb7aad2c	delivered	2018-02-13 21:18:39	2018-02-13 22:2
a4591c265e18c1d0ee52889e2d8acc3	503740e9ca751c0dda7ba29e9ab8f608	delivered	2017-07-09 21:57:05	2017-07-09 22:1
136coe7faa42f2d0ce453f5dc79a6098	ed0271e0b7da060a393796590e7b737a	invoice	2017-04-11 12:22:08	2017-04-13 13:2

99,441 orders were placed in total.

Question 2: How many orders were delivered successfully?

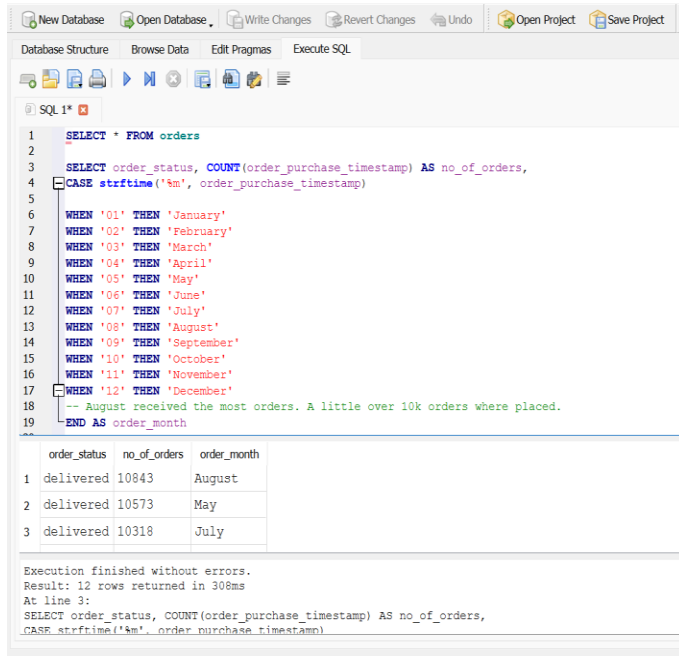


The screenshot shows a database management tool with a SQL editor and a results pane. The SQL editor contains the query: `SELECT COUNT(*) AS successful_delivery FROM orders WHERE order_status = 'delivered'`. The results pane displays a table with 1 column: `successful_delivery`. The table contains 1 row of data, representing the count of successfully delivered orders.

successful_delivery
96478

96,478 out of 99,441 orders were successfully delivered, with over 95% delivery success rate.

Question 3: Which month had the most orders?



The screenshot shows a SQL IDE with a query editor and a results pane. The query is as follows:

```
1 SELECT * FROM orders
2
3 SELECT order_status, COUNT(order_purchase_timestamp) AS no_of_orders,
4 CASE strftime('%m', order_purchase_timestamp)
5
6 WHEN '01' THEN 'January'
7 WHEN '02' THEN 'February'
8 WHEN '03' THEN 'March'
9 WHEN '04' THEN 'April'
10 WHEN '05' THEN 'May'
11 WHEN '06' THEN 'June'
12 WHEN '07' THEN 'July'
13 WHEN '08' THEN 'August'
14 WHEN '09' THEN 'September'
15 WHEN '10' THEN 'October'
16 WHEN '11' THEN 'November'
17 WHEN '12' THEN 'December'
18 -- August received the most orders. A little over 10k orders were placed.
19 END AS order_month
```

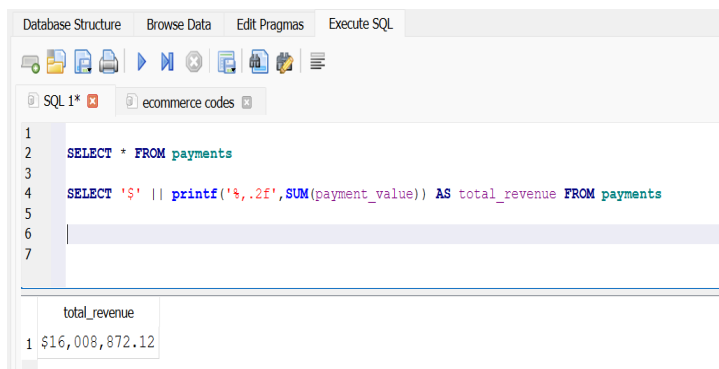
The results pane shows the following data:

	order_status	no_of_orders	order_month
1	delivered	10843	August
2	delivered	10573	May
3	delivered	10318	July

Execution finished without errors.
Result: 12 rows returned in 308ms
At line 3:
SELECT order_status, COUNT(order_purchase_timestamp) AS no_of_orders,
CASE strftime('%m', order_purchase_timestamp)

August was the busiest month with 10,843 orders, followed by May with 10,573 and July with 10,318. The top 3 months are all mid-year, suggesting a strong sales peak in the middle of the year.

Question 4: What is the total revenue generated?



The screenshot shows a SQL IDE with a query editor and a results pane. The query is as follows:

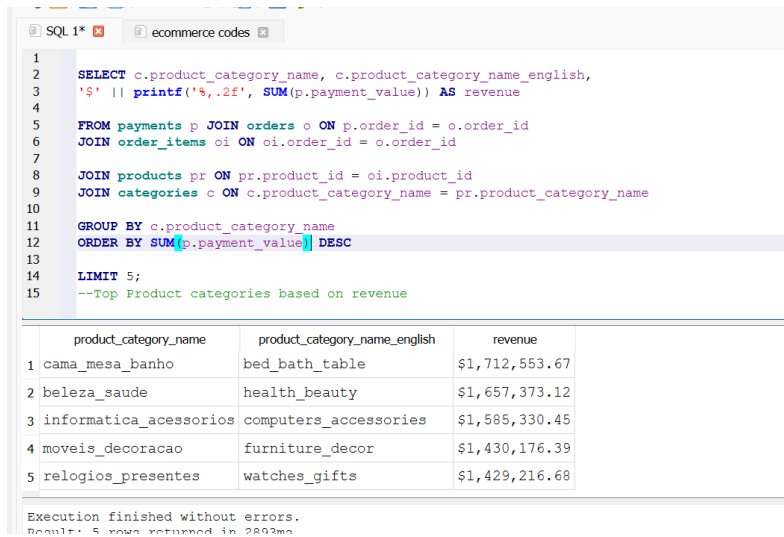
```
1
2 SELECT * FROM payments
3
4 SELECT '$' || printf('%,.2f', SUM(payment_value)) AS total_revenue FROM payments
5
6
7
```

The results pane shows the following data:

	total_revenue
1	\$16,008,872.12

*Total revenue generated was **\$16,008,872.12**, over 16 million dollars across 99,441 orders, giving an average order value of roughly \$161*

Question 5: What are the top 5 product categories by revenue?



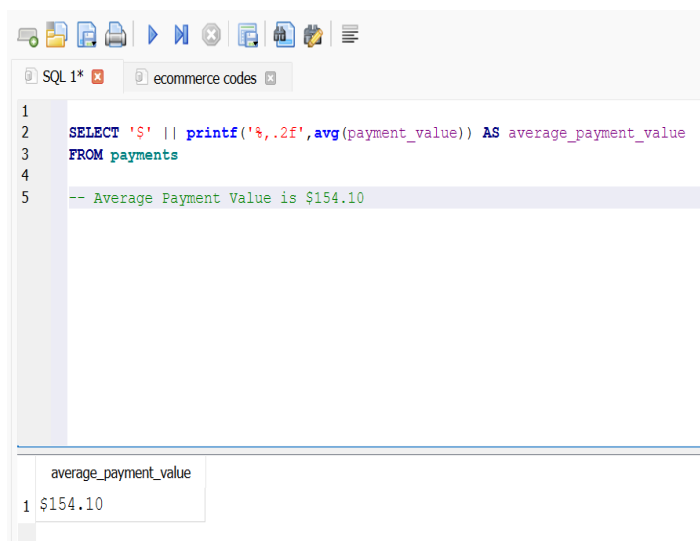
```
1
2 SELECT c.product_category_name, c.product_category_name_english,
3 '$' || printf('%.2f', SUM(p.payment_value)) AS revenue
4
5 FROM payments p JOIN orders o ON p.order_id = o.order_id
6 JOIN order_items oi ON oi.order_id = o.order_id
7
8 JOIN products pr ON pr.product_id = oi.product_id
9 JOIN categories c ON c.product_category_name = pr.product_category_name
10
11 GROUP BY c.product_category_name
12 ORDER BY SUM(p.payment_value) DESC
13
14 LIMIT 5;
15 --Top Product categories based on revenue
```

	product_category_name	product_category_name_english	revenue
1	cama_mesa_banho	bed_bath_table	\$1,712,553.67
2	beleza_saude	health_beauty	\$1,657,373.12
3	informatica_acessorios	computers_accessories	\$1,585,330.45
4	moveis_decoracao	furniture_decor	\$1,430,176.39
5	relogios_presentes	watches_gifts	\$1,429,216.68

Execution finished without errors.
Result: 5 rows returned in 2853ms

Bed, bath, and table was the top revenue category at \$1,712,553.67, followed by health and beauty at \$1,657,373.12 and computer accessories at \$1,585,330.45.

Question 6: What is the average payment value per order?



```
1
2 SELECT '$' || printf('%.2f', avg(payment_value)) AS average_payment_value
3 FROM payments
4
5 -- Average Payment Value is $154.10
```

	average_payment_value
1	\$154.10

The average payment value per order was \$154.10.

Question 7: How many orders had bad reviews (score 1 or 2)?

The screenshot shows a SQL IDE with a query editor and a results pane. The query editor contains the following SQL code:

```
1 SELECT COUNT(*) FROM order_reviews
2 SELECT COUNT(order_id) AS count_bad_reviews
3 FROM order_reviews
4 WHERE review_score <=2;
5
6 -- 14,575 orders had low review tags, accounting for less than 19% of overall orders reviewed.
```

The results pane shows a single row with the column name 'count_bad_reviews' and the value '14575'.

count_bad_reviews
14575

14,575 orders received bad reviews, below 19% of all orders, a meaningful dissatisfaction rate any business would want to investigate.

Question 8: Which sellers have the most orders?

The screenshot shows a SQL IDE with a query editor and a results pane. The query editor contains the following SQL code:

```
1 --sellers with the most orders
2
3
4 SELECT seller_id, COUNT(order_id) AS total_orders
5 FROM order_items
6 GROUP BY seller_id
7 ORDER BY total_orders DESC
8
9 -- the table displays the top five sellers by total orders. Each received
10 -- at least 1.5k orders.
```

The results pane shows a table with two columns: 'seller_id' and 'total_orders'. The table contains five rows of data.

	seller_id	total_orders
1	6560211a19b47992c3666cc44a7e94c0	2033
2	4a3ca9315b744ce9f8e9374361493884	1987
3	1f50f920176fa81dab994f9023523100	1931
4	cc419e0650a3c5ba77189a1882b7556a	1775
5	da8622b14eb17ae2831f4ac5b9dab84a	1551

Execution finished without errors.

The top seller processed 2,033 orders. All 5 top sellers handled at least 1,500 orders each.